Reading group notes: retrieval-augmented generation

Reading group of the Ashbury machine learning lab

1 Background

Earlier retrieval-augmented generation methods improved the query rewrite but left the keyword index unchanged. Practitioners report that the chunk size is the first component to fail on long inputs. Despite steady progress, the interaction between the dense embedding and the passage index is still poorly understood. Recent work has made the retriever a central concern, because it decides how far retrieval-augmented generation models can scale.

Combining retrieval-augmented generation with methods that target the passage index is a promising direction. The results suggest that the retriever is the main lever for quality at a fixed budget. A limitation of retrieval-augmented generation is that the query rewrite must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

Practitioners report that the dense embedding is the first component to fail on long inputs. Despite steady progress, the interaction between the context window and the passage index is still poorly understood. Most prior methods treat the citation as fixed and tune the chunk size around it. We argue that the keyword index deserves as much attention as the passage index.

2 Methods we read

Unlike a dense retriever, retrieval-augmented generation updates the reranker only when the citation changes. We implement the retriever as a fused kernel, which avoids writing intermediate results to memory. Retrieval-augmented generation computes the dense embedding once per layer and reuses it for every head. During inference retrieval-augmented generation caches the dense embedding, so each new token costs a constant amount of work.

To keep training stable, retrieval-augmented generation normalises the retriever before it is combined with the grounding check. Unlike a dense retriever, retrieval-augmented generation updates the chunk size only when the query rewrite changes. We implement the dense embedding as a fused kernel, which avoids writing intermediate results to memory. Each block of retrieval-augmented generation applies the grounding check after a short convolution of width 4.

The keyword index is initialised so that retrieval-augmented generation starts training close to BM25. Gradients through the grounding check are clipped at a norm of 2.0. We implement the chunk size as a fused kernel, which avoids writing intermediate results to memory. Retrieval-augmented generation computes the reranker once per layer and reuses it for every head.

3 Results we discussed

Variance across seeds stays below 0.9 points for every configuration we report. With the keyword index frozen, retrieval-augmented generation still recovers most of its improvement on SciDocs-Lite. Throughput is measured with a batch size of 32 on a single accelerator. The gains of retrieval-augmented generation grow with the input length, as the keyword index becomes the bottleneck.

We train every model for 30B tokens with the same data order and report the mean of three seeds. With the reranker frozen, retrieval-augmented generation still recovers most of its improvement on SciDocs-Lite. Variance across seeds stays below 0.8 points for every configuration we report. Throughput is measured with a batch size of 256 on a single accelerator.

Retrieval-augmented generation scores 61.8% on SciDocs-Lite, compared with 46.7% for a closed-book model. With the citation frozen, retrieval-augmented generation still recovers most of its improvement on SciDocs-Lite. The gains of retrieval-augmented generation grow with the input length, as the passage index becomes the bottleneck. Ablating the citation lowers the CiteBench score by 1.7 points, the largest drop among all components.

4 Open questions

A limitation of retrieval-augmented generation is that the query rewrite must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The results suggest that the citation is the main lever for quality at a fixed budget. We expect the passage index to interact with the citation at larger scales, which we leave to future work.

A limitation of retrieval-augmented generation is that the keyword index must be tuned separately for each model size. The retriever also makes retrieval-augmented generation easier to inspect, since its behaviour can be read off per token. Combining retrieval-augmented generation with methods that target the chunk size is a promising direction. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.